

# MPG No. 11.2.36 $4/m\bar{1}'$ [ Type II, tetragonal ] [G tensor]

\* Rank 0 tensor. \* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & G_z \end{bmatrix}$$

$$G_z = G_{pz}^{(1)}$$

\* Rank 2 tensor (s). \* Rank 2 tensor (a). \* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & G_{zzz} \\ 0 & G_{xxz} & 0 \\ G_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = -G_{faz}^{(1)} + G_{pz}^{(1)}$$

$$G_{zzz} = 2G_{faz}^{(1)} + 3G_{pz}^{(1)}$$

\* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & 0 \\ G_{yzx} & -\frac{G_{xxz}}{2} & 0 \\ -\frac{G_{xxz}}{2} & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = 2G_{pz}^{(2)}$$

$$G_{yzx} = 3Q_{du}^{(1)}$$

$$\frac{G_{xxz}}{2} = G_{pz}^{(2)}$$

\* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & 0 \\ -G_{yzy} & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_s^{(1)}$$

$$G_{yzy} = G_{pz}^{(3)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

\* Rank 4 tensor (sss). \* Rank 4 tensor (ssa). \* Rank 4 tensor (aas). \* Rank 4 tensor (aaa). \* Rank 4 tensor (sa). \* Rank 4 tensor (as). \* Rank 4 tensor (s). \* Rank 4 tensor (a). \* Rank 4 tensor (t).